BENJAMIN BARLOG

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EDUCATION

University of Montana, Missoula, MT — Expected May 2027

B.S. Computer Science, Data Science concentration; Minor in Mathematics · GPA: 3.61 (institution)

Relevant coursework: Data Structures & Algorithms, Operating Systems, Computer Architecture, Probability & Simulation, Linear Algebra, Discrete Mathematics, Data Visualization, Geospatial Data Science

RESEARCH EXPERIENCE

Undergraduate Research Assistant, AI in Education — University of Montana — Jan 2026 – Present

Mentor: Dr. Zedong Peng, Department of Computer Science

Studied whether LLM tutors give responses that suit a learner's knowledge and place in a course, not just correct ones. Developed the Pedagogical Suitability Index (PSI), a composite metric for scoring the pedagogical fit of tutoring responses, and used it to evaluate and improve the output of four LLMs.

- Designed the PSI evaluation metrics: temporal violation, knowledge distance, cognitive load, and Bloom's taxonomy alignment
- Applied PSI across 240 evaluations of four LLMs; PSI-guided regeneration, built by collaborators, improved 51 of 62 low-scoring responses (82.3%)
- First author; presented the paper at IEEE IRI 2026 in Seattle, WA

Terrain Masking and Least-Exposure Routing (independent, in progress) — University of Montana — Oct 2026 – Present

Mentor: Dr. Jesse Johnson, Data Visualization course

Testing whether least-exposure routes for low-altitude drones under notional line-of-sight sensors favor lower, more rugged terrain than shortest routes, and how that changes with altitude. Sole author of the hypothesis, method, and implementation.

- Building a pipeline on USGS 3DEP elevation data: per-sensor minimum-visible-altitude rasters, probabilistic detection, and a survival-based path cost ($-\ln(1-P)$) solved exactly with Dijkstra's algorithm
- Designed a paired statistical test (Wilcoxon signed-rank) across randomized sensor layouts, start–goal pairs, and three flight altitudes

Housing Affordability in Recreation-Amenity Counties (independent) — University of Montana — Aug – Sep 2026

Mentor: Dr. Jesse Johnson, Geospatial Data Science course

Tested whether counties with strong outdoor-recreation access and little high-wage industry are unaffordable for local workers, across 2,753 U.S. counties.

- Built a home-price-to-local-wage metric (Zillow, BLS QCEW) and a recreation amenity index from federal-land polygon overlays and recreation employment (GeoPandas, statsmodels, QGIS)
- Found a significant amenity × industry interaction ($p < 10^{-13}$): amenity's effect on unaffordability was ~3× larger where high-wage industry was scarce, acting through prices rather than wages
- Caught and corrected a boundary-data error that overstated eastern national-forest land by up to 55 percentage points

Personacue.ai — Co-founder & Lead Backend Developer — Mar 2026 – Present

Mentor: Dr. Zedong Peng, University of Montana; with Hudson Craig (lead frontend)

LLM-based platform that replicates the personalities of historical and fictional figures on locally hosted models. Details are limited by a non-disclosure agreement.

- Lead backend development; co-developed the original retrieval-augmented generation (RAG) pipeline for aligning local models with personas, roles, and knowledge bases.

PUBLICATIONS & PRESENTATIONS

B. Barlog, H. Craig, and Z. Peng, "Evaluating and Improving Pedagogical Fit in LLM-Based AI Tutors with the Pedagogical Suitability Index," in *Proc. IEEE 27th Int. Conf. on Information Reuse and Integration for Data Science (IRI)*, Seattle, WA, USA, 2026. arXiv:2608.05411.

- Paper presentation, IEEE IRI 2026, Seattle, WA, Jul. 31 – Aug. 2, 2026

TECHNICAL PROJECTS

Ultrasonic Scanning Sonar — Arduino, C++, Python Sep 2026

- Built a servo-swept ultrasonic scanner (Arduino Uno, HC-SR04) that streams angle–distance readings over serial to a live polar plot, mapping a 180° slice of a room
- Wrote non-blocking servo timing and 6-sample per-angle averaging to reduce noise; detected a test object at 25 cm

Swarm Intelligence Self-Study — Python Aug 2026

- Directed an AI-assisted build of an interactive 2D drone-swarm simulator (boids flocking, noisy range-limited sensing, Kalman filtering) and used it to study how individual parameters and sensor noise affect swarm cohesion

Data Visualization Studies — Observable 2026

- Coastal proximity and fitness: coastal counties show more fitness-related activity, likely reflecting affluence rather than the coastline itself
- Campaign finance and wealth inequality: sparse donation data supported at most a weak positive correlation

Investment Comparison Model Mar 2024

- Derived a multi-parameter model of the relative net value of two investment strategies over time; analyzed its derivatives to find which market conditions favor each

PROFESSIONAL EXPERIENCE

First Interstate Bank — Database Administrator Intern, Billings, MT (remote) Jun 2025 – Sep 2025

- Developed an in-house server-monitoring solution to replace a third-party system, saving roughly $70,000 per year
- Supported DBAs with server health monitoring, incident reporting, deadlock troubleshooting, and query optimization (SQL Server, SSMS, SolarWinds DPA, Power BI)

First Interstate Bank — Wealth Management Admin. Assistant, Billings, MT July 2022 – Dec 2022

- Assisting in scheduling appointments, making transactions, and
- Prepared and delivered presentations on current economic conditions to colleagues

SKILLS

Languages: Python, C/C++, Java, SQL
Data & ML: pandas, NumPy, GeoPandas, statsmodels, matplotlib, QGIS, Observable, Power BI
Tools: Git, PlatformIO/Arduino, SQL Server (SSMS)
Interests: economics, defense autonomy, fitness, investing, poker, cooking